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# PAM SI Cloud

Virtual Workshop

March 2026

Presenters: Rebecca Van Hoeck (NEFSC)

PAM SI Lead: Sofie Van Parijs (NEFSC)

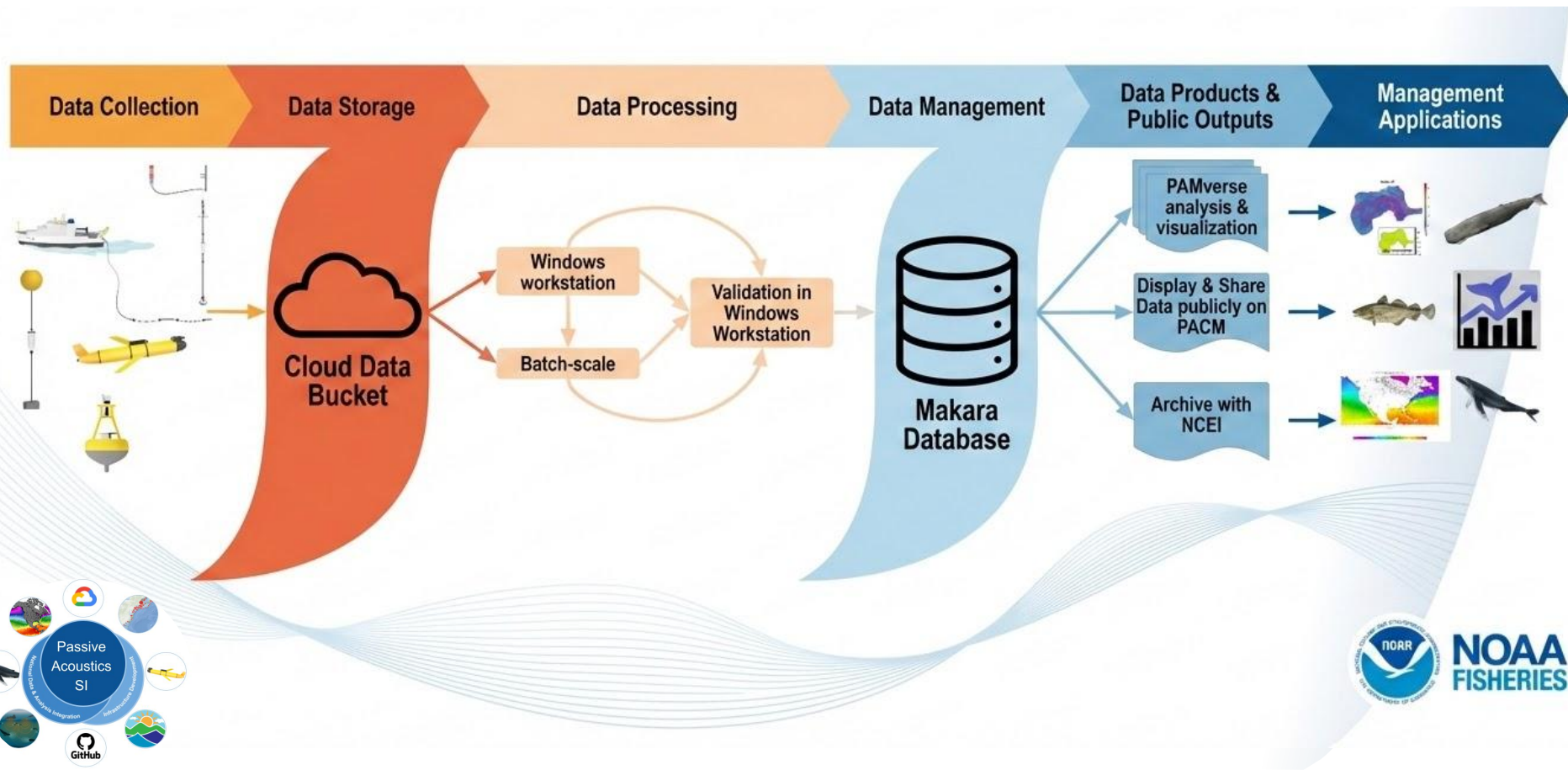
Developers: Dan Woodrich (AFSC), Jordan Shimer, Devin Mullin, Jon Windmiller, Hannan Khan (TSPI/OCIO)

# Challenges with on-premise capability

| Challenge                                                                                                                   |   | Impact                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>● Reliant on science center network drives or external hard drives</li></ul>          | → | <ul style="list-style-type: none"><li>● Difficult data sharing and archiving, limiting national collaboration</li></ul>              |
| <ul style="list-style-type: none"><li>● Due to data volume, PAM programs often acquired on-premise infrastructure</li></ul> | → | <ul style="list-style-type: none"><li>● Expensive, redundant infrastructure costs to programs</li></ul>                              |
| <ul style="list-style-type: none"><li>● Limited, shared computing resources constrained to sequential processing</li></ul>  | → | <ul style="list-style-type: none"><li>● Time spent waiting for processing jobs to finish or resolving server interruptions</li></ul> |

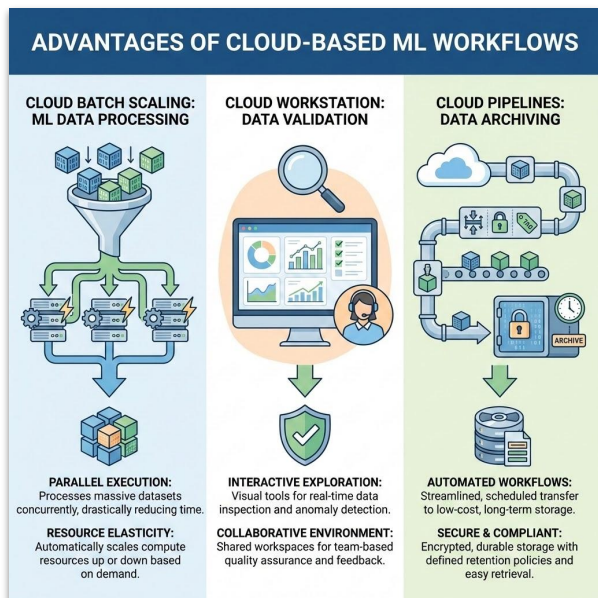


# Creating standardized, cloud-based, and open source workflows for PAM data

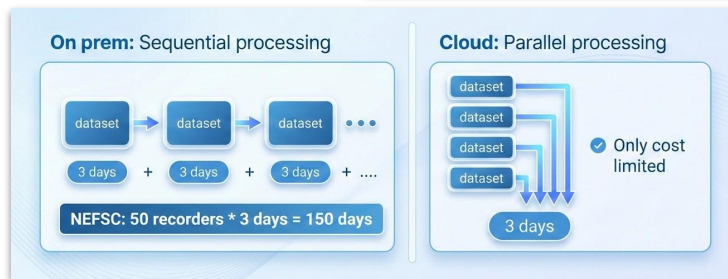


# Modernized PAM data workflows in the cloud

Cloud migration has provided the ability to hyper-scale data processing and streamline data archiving while maintaining a familiar workflow. This removes regional IT constraints, streamlines data transfers, and reduces the time to go from raw data to informative, publicly available data products.



*Developing a Windows Workstation and collaboration with NMFS OCIO has been key to our cloud success*



# National PAM Cloud Overview

## Data Storage:

- Archiving pipeline
  - Quincy Cantu, Clint Lohr, Hannan Khan, Jeff Walker (CIRES/NCEI, TSPI Ltd, NEFSC/Walkerenvres Ltd)
- Cloud Storage
  - Dan Woodrich (CICOES/AFSC)
- Cloud-based Navy review
  - Jason Gedamke (OST)

## Cloud Processing:

- Windows workstations
  - Dan Woodrich (CICOES/AKFSC)
- Batch-scale processing:
  - Becca Van Hoeck, Genevieve Davis, Ann Allen, Dan Woodrich (NEFSC, PIFSC, CICOES/AKFSC)

**Learn more on the PAM  
Cloud GitHub!**

[https://nmfs-ost.github.io/  
PAM-Cloud/](https://nmfs-ost.github.io/PAM-Cloud/)



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# Cloud Processing

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# Leveraging Google Cloud Composer

Single user interface orchestrating passive acoustic detectors at scale



## 10+

**Acoustic Processes**  
Single UI Access

### Unified Workflow Orchestration

Centralizes control for over 8 specialized passive acoustic detectors through a streamlined Google Cloud Composer interface, simplifying operational complexity and accelerating analytical workflows.



### Native Cloud Integration

Direct integration with cloud storage for seamless data ingest, management, and efficient retrieval across pipelines.



### Makara DB Automation

Automated processing triggers and metadata handling built directly on top of the Makara database ecosystem.



### Parallel Scaling

Flexible compute resource allocation allowing massive parallel processing of high-volume acoustic datasets.



### Cost Efficiency

Optimized resource utilization ensuring high-throughput detection pipelines remain cost-effective at scale.

# PAM Cloud Workflow Efficiency

Accelerating passive acoustic monitoring through cloud migration and parallel processing



up to **95%**  
Processing Time Savings

## Transition to High-Scale Parallel Processing

Migrating from legacy sequential desktop processing to scalable cloud infrastructure enables simultaneous multi-dataset execution, eliminating severe analytical bottlenecks.



### Parallel Scale

Replaces sequential, single-machine runs with automated, cloud clusters to process high-volume acoustic data concurrently.



### Metadata Automation

Automates pipeline configuration and metadata extraction, significantly reducing manual overhead and configuration setup time.



### Analyst Focus

Frees analyst capacity from routine processing tasks, shifting focus toward high-value research, data interpretation, and analytical outputs.

# Cloud Composer Demo

- **Multi-species machine learning model**
- **SoundChecker: AFSC spectrogram generation**
- **Low Frequency Detection & Classification System (LFDCS)**
  - Trained on North Atlantic call types for North Atlantic right, humpback, sei, fin, and blue whales
- **Hybrid millidecade Soundscape Metrics via PyPAM-based Processing**
- Minke and Humpback CNN detector
- INSTINCT Inference
- NCEI Archiving sub
- Perch/Hoplite Model Framework (*in development*)
- Audio file conversion and decimation (*in development*)
- PAM Glider Processing (*up next*)

**What are our remaining development needs?**



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# Multispecies Machine Learning Model

Automated bioacoustic classification pipeline for marine mammal detection

## 8 Classes

### Trained with Google

Using acoustic data from diverse sources

Provides a score for each class

### Pipeline & Infrastructure

#### Workflow

Streamlined to run batch processing directly on your cloud data

Airflow: [Multispecies\\_ML\\_whale\\_model](#)

#### Data Output

BigQuery Database- timestamps and 'scores'

#### Caveats:

This is a static model trained on several different data types for discrete call types.

### Target Detection Classes

#### Blue Whale

Central Pacific Call

#### Fin Whale

20 Hz Call

#### NARW

Upcall

#### NPRW

Upcall

#### NPRW

Gunshot

#### Bryde's Whale

Biotwang

#### Orca

Whistles

#### Minke Whale

Boing

Model: <https://www.kaggle.com/models/google/multispecies-whale>

Blog with more info: <https://research.google/blog/whistles-songs-boings-and-biotwangs-recognizing-whale-vocalizations-with-ai/>



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# Multispecies Machine Learning Model

Automated bioacoustic classification pipeline for marine mammal detection

Trigger DAG: multispecies\_ML\_whale\_model

DAG conf Parameters

instances \*:

can be either a list of files or directories

bq\_table\_name \*:

location for BQ output

upload\_images:



dry\_run:



Generated Configuration JSON and Dagrun Options ▾

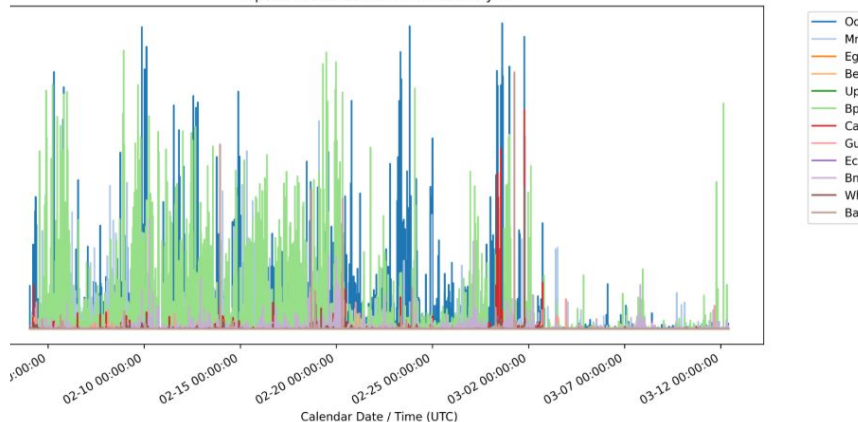
Trigger

Cancel

## Pipeline Outputs

- Structured export to **BigQuery database** (timestamps & confidence scores)
- Automated figure generation for each audio file: **Logits & Probabilities**

Species Probabilities Timeline Analysis



Model: <https://www.kaggle.com/models/google/multispecies-whale>

Blog with more info: <https://research.google/blog/whistles-songs-boings-and-biotwangs-recognizing-whale-vocalizations-with-ai/>

# SoundChecker pngs

Pregenerated spectrograms at different contrast levels for fast paging with SoundChecker (manual analysis)

## Trigger DAG: wav2png\_cloud\_batch\_orchestrator

Select Recent Configurations

Default parameters

### DAG conf Parameters

config\_gcs\_uri \*:

gs://pam-ww-tmp/Daniel\_Woodrich/config.json

Full GCS URI to the configuration JSON.

mooring\_name \*:

AL18\_AU\_PH01

Name of the mooring (e.g., AL18\_AU\_PH01)

freq\_code \*:

3

Select the frequency code. 1 = LOW, 2 = REG, 3 = SHI, 4 = SPL

processing\_style \*:

cheap

Opt for 'cheap' if you are ok with this potential taking 6-12 hours,

target\_month:

(Optional) Specific month to process, e.g., '01\_2019'. This is only

## Pipeline Outputs

- Spectrograms in a view folder for quality assurance, OR
- Spectrograms in the app expected folder for general use

